

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
01.2	• The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	
02.1	• There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient	2

Question	Marking guidance	Marks
02.2	• Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises • Calculate standard deviation to assess scatter: more variable data need more samples • Test the method (extraction technique, timing) for reliability before the main study	4
02.3	• Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates • Collect worms that come to the surface for a standardised time; count / identify / weigh • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
03.1	• Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant • Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate • Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers • Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor • Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away • Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions	6
03.2	• The organisation / person that causes the pollution is responsible for (paying for) the damage caused • If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely	2
03.3	• It is assumed that the waste will cause pollution / harm until research shows it is unlikely to • So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
04.1	• Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines • Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost	2

Question	Marking guidance	Marks
04.2	• As volume increases the surface area : volume ratio decreases • Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage)	2
04.3	• Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation • Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines	2
04.4	• Materials with high specific heat capacity (concrete, water) warm and cool slowly • They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating	2
04.5	<i>Any 2 from:</i> • Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation • Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage • Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	
05.1	<i>Any 3 from:</i> • Unburnt fuel in vehicle exhaust gases • Evaporation from fuel tanks / spillage at filling stations • Solvents (paints, cleaning products) • Aerosol propellants • Gas leaks / fossil fuel exploitation (methane)	3
05.2	• Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon	2
05.3	• A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) • NO_x are chemically reduced to nitrogen and oxygen	3
05.4	• Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

Question	Marking guidance	Marks
06.1	• Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships • Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption • Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again	6
06.2	• Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) • Steel wheels do not deform so rolling resistance is very low	2
06.3	<i>Any 2 from:</i> • More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel • Better temperature control (thermostatic fan) • Electronic ignition control / regular servicing to keep spark timing correct • Turbochargers supplying more oxygen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
07.1	• Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic • Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it; the cold water returns southwards at depth (thermohaline circulation)	2
07.2	• Higher temperatures melt land ice on Greenland which flows into the North Atlantic • The fresh water dilutes the seawater, reducing its salinity and density • Less dense water sinks less easily so the flow rate of the current is reduced • Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise	4
07.3	• Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them • Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change; individual events do not show trends • Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity); incomplete understanding of the processes	3
07.4	• Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data; the sequence of positions shows current direction and speed (surface currents also by satellites and buoys)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	

Question	Marking guidance	Marks
08.1	• $110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0$ • $= 44.8 \text{ t ha}^{-1} \text{ yr}^{-1}$ (accept 44.7-44.8)	2
08.2	• $110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25$ • $= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}$	2
08.3	• Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall • Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable • Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land ($C = 0.02$) needed to bring loss below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$	3
08.4	• Factors are averages / arbitrary values for typical conditions; rainfall erosivity varies between years / storms • Does not include gully erosion, landslides, wind erosion or local variation in soil; based on data from particular regions	2
08.5	• C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	
09.1	• The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling	2
09.2	• Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced)	2
09.3	• Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
10.2	<i>Any 1 from:</i> • Biological control (introduced predators / pathogens) • Predator habitats (hedgerows, beetle banks) • Crop rotation • Companion / barrier crops • Sterile male technique / pheromone traps • Genetic resistance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
11.1	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25